

Seminar 5 and 6 - Exercises

1. Let's assume that a firm has in the long run production function $LTC = 2y^3 - 5y^2 + 15y$
 - Find long run average costs (LAC) and long run marginal costs (LMC).
A: $LAC = 2y^2 - 5y + 15$, $LMC = 6y^2 - 10y + 15$
 - Find the amount of y when positive economies of scales turn into negative. (Minimum LAC)
A: $y = \frac{5}{4}$
 - Calculate when the shape of LTC turns from concave to convex. (Minimum LMC)
A: $y = \frac{5}{6}$
 - Draw all obtained values (minimum LAC , minimum LMC etc.) into one chart.
A: See seminar
2. A firm has cost function $STC(y) = 2y^2 - 50y + 40$. Assume perfect competition market.
 - Calculate VC, FC, AVC, AFC and SMC?
A: $VC = 2y^2 - 50y$, $FC = 40$, $AVC = 2y - 50$, $AFC = \frac{40}{y}$, $SMC = 4y - 50$
 - In which point have SAC its minimum?
A: $y = \sqrt{20}$
 - Draw a chart with all those functions.
A: see seminars
3. Short term production function of a firm is $f(x_1, x_2) = (x_1 + x_2^2)^{\frac{1}{2}} + 5$ where the second input x_2 is an investment into a new machinery needed for production of a product. Therefore this input is fixed an equal to 4 in the short term.
 - Derive the short production function $f(x_1, 4)$. For the required amount of product $y = 10$ calculate the optimal level of input x_1 .
A: $f(x_1, x_2) = (x_1 + 16)^{\frac{1}{2}} + 5$, $x_1 = 9$
 - Derive the cost function $C(w, y)$ and prove that the Shepard's lemma holds.
A: $C = w_1(y - 5)^2 - 16w_1 + 4w_2$, yes
4. Otto sells cars. His brother Dent is in the auto repair business. Dent recently had little else to do and decided to calculate his cost conditions. He found that the total cost of repairing s cars is $c(s) = 2s^2 + 10$. But Dent's attention was diverted to other things ...
 - Dent's Total Variable Costs ? A: $2s^2$
 - Total Fixed Costs? A: 10
 - Average Variable Costs? A: $2s$
 - Average Fixed Costs? A: $10/s$
 - Average Total Costs? A: $2s + 10/s$
 - Marginal Costs? A: $4s$
5. Mary wants to open a flower shop in a new mall. She has her choice of three different floor sizes: 200m², 500m², or 1000m². The rent will be \$1 per m². Mary estimates that if she has F m² of floor space and sells y bouquets a month, her variable costs will be $c_v(y) = y^2/F$ per month. Recalculate for all three possibilities.
 - What is her $MC(y)$, $AC(y)$?
A: **200**: $\frac{y}{100}$, $\frac{200}{y} + \frac{y}{200}$; **500**: $\frac{y}{250}$, $\frac{500}{y} + \frac{y}{500}$; **1000**: $\frac{y}{500}$, $\frac{1000}{y} + \frac{y}{1000}$
 - At what amount of output is $AC(y)$ minimized? What is the average costs at this level of output?
A: 200; 500; 1000; AC is always 2 at given levels of output.
 - Use chart to show Mary's marginal cost curve is. Then show her average cost curve using points where y is 100, 200, 500 and 1000.
A: graph is provided in the setting of this seminar.

6. Firm has a production function $f(x_1, x_2, x_3) = (x_1^{\frac{1}{2}} + x_2^{\frac{1}{2}})^2 + 4x_3$, where x_1 represents number of hours of labour, x_2 number of machinery hours and x_3 material. Firm is able to use machinery maximally for 81 hours in the short-run and has already paid for every machinery hour $w_2 = 30$

- Derive short term production function $(x_1, 81, x_3)$. Calculate short-run conditional demands for labour and material.

A: $f(x_1, 81, x_3) = (x_1^{\frac{1}{2}} + 9)^2 + 4x_3$

Conditional demands: $x_1 = \left(\frac{9w_3}{4w_1 - w_3}\right)^2$, $x_2 = 81$, $x_3 = \frac{1}{4} \left[y - \left(\frac{36w_1}{4w_1 - w_3}\right)^2 \right]$

- Derive cost function $c(w, y, 81)$

A: $C(w, y, 81) = w_1 \cdot \left(\frac{9w_3}{4w_1 - w_3}\right)^2 + 81 \cdot 30 + w_3 \cdot \frac{1}{4} \left[y - \left(\frac{36w_1}{4w_1 - w_3}\right)^2 \right]$

7. Consider a firm with two plants. Assume that the plants have following short-term cost functions:

$$\begin{aligned} STC_1 &= 2y^2 + 600 \\ STC_2 &= y^2 + 1000 \end{aligned}$$

- Calculate $SAC_1(y)$, $AVC_1(y)$ and $SMC_1(y)$ and $SAC_2(y)$, $AVC_2(y)$ and $SMC_2(y)$.

A: $SAC_1(y) = 2y + \frac{600}{y}$, $AVC_1(y) = 2y$, $SMC_1(y) = 4y$

$SAC_2(y) = 2y + \frac{1000}{y}$, $AVC_2(y) = y$, $SMC_2(y) = 2y$

- What proportion would a firm produce in plant 1 and what proportion in plant 2?

A: $SMC_1 = SMC_2 \dots 2y_1 = y_2$; it means that the firm will produce twice as much in the second plant.

8. Consider a firm with two plants. Assume that the plants have following short-term cost functions:

$$\begin{aligned} STC_1 &= 3y^2 + y + 504 \\ STC_2 &= 2y^2 + y + 700 \end{aligned}$$

- Calculate $SAC_1(y)$, $AVC_1(y)$ and $SMC_1(y)$ and $SAC_2(y)$, $AVC_2(y)$ and $SMC_2(y)$.

A: $SAC_1(y) = 3y + 1 + \frac{504}{y}$, $AVC_1(y) = 3y + 1$, $SMC_1(y) = 6y + 1$

$SAC_2(y) = 2y + 1 + \frac{700}{y}$, $AVC_2(y) = 2y + 1$, $SMC_2(y) = 4y + 1$

- Calculate elasticity of cost with respect to output e_y^c for both plants and determine the interval where STC_1 and STC_2 have negative economies of scale.

A: Plant 1: $e_{y_1}^c = \frac{6y_1 + 1}{3y_1 + 1 + \frac{504}{y_1}} = \frac{6y_1^2 + y_1}{3y_1^2 + y_1 + 504}$.

The interval where STC_1 have negative economies of scale is where $e_{y_1}^c > 1$: $y_1 \in [\sqrt{168}, \infty]$.

Plant 2: $e_{y_2}^c = \frac{4y_2 + 1}{2y_2 + 1 + \frac{700}{y_2}} = \frac{4y_2^2 + y_2}{2y_2^2 + y_2 + 700}$;

STC_2 have negative economies of scale where $y_2 \in [\sqrt{350}, \infty]$.

- What proportion would a firm produce in plant 1 and what proportion in plant 2?

A: $\frac{3}{2}y_1 = y_2$